Screen-free game kit · Design; not tested in practice

A Drop's Journey



Learning goal

Set up an experiment that compares bare and covered soil fairly: only the cover changes, everything else stays the same. Keep the prediction separate from the result.

Materials

The card pages of this PDF (A4, 100% scale), scissors, pencils; box labels go on the table.

How to play



- No time limit; nobody is rushed.
- A wrong answer costs nothing: read why, then try again.
- A team that wants help turns over İKO's hint card.
- At the end, say what you learned in one sentence and choose the real-life task.

Task flow

Ages 7–9

İKO will make rain on two trays. First let us set up a fair experiment, then make a prediction.

1. Which things must be the SAME on both trays? Choose all.
2. Your prediction: in which cup will more water collect?
3. Illustrative animation step: in the screen-free version the teacher reads this text. *(ILLUSTRATIVE ANIMATION: rain falls on both trays at the same time. This animation does not show the real result; we left the cups empty. We will measure the result in the class experiment.)*
4. If your prediction does not match after the experiment, what do you do?

Ages 10–12

Name the variables and judge whether two teams' experiments are fair.

1. Place each item in the right type of variable.
2. Team A poured the water on the bare tray in 15 seconds and on the covered tray in 60 seconds. Is this comparison fair?
3. Team B pressed down the soil on the covered tray but not on the bare tray. What do you say?
4. Whatever the class result is, which sentence limits the conclusion correctly?



Real-life task: With your teacher's permission, after rain, look at a patch of bare soil and a patch covered with leaves in the garden. Draw the difference. If you do not want to touch the soil, just look.

If a badge is used, it only shows the task was finished; it is not proof of learning.

A Drop's Journey · Ages 7–9 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 7–9 · Question card · ☐☐

Which things must be the SAME on both trays? Choose all.

A. Amount of soil
B. Amount of water (100 mL)
C. Slope of the tray
D. Pouring the water at the same time
E. The cover on top of the soil
F. Waiting time (3 minutes)

Ages 7–9 · Hint



Only one thing will be different: the cover on top of the soil.

Ages 7–9 · Question card

Your prediction: in which cup will more water collect?

A. In the bare soil cup
B. In the covered soil cup
C. The same

Ages 7–9 · Hint



A prediction is not wrong; it is an idea.

Ages 7–9 · Question card

If your prediction does not match after the experiment, what do you do?

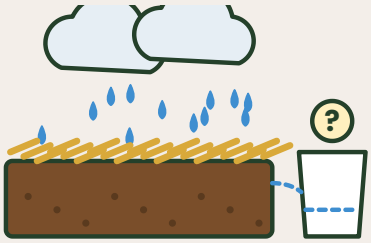
A. That is fine. I write my prediction and the result separately.
B. I change the result to match my prediction.

Ages 7–9 · Hint

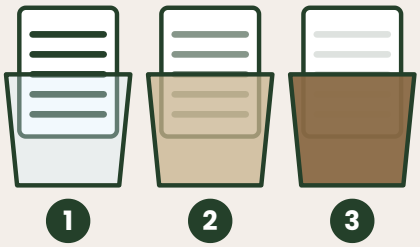


Scientists make predictions too; when the result is different, they write it down.

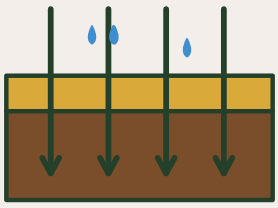
A Drop's Journey · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm



What we change



What we measure



What we keep the same

Ages 10–12 · Cut-out cards

Cover (bare / dry straw)

Ages 10–12 · Cut-out cards

Water collected in the cup (mL)

Ages 10–12 · Cut-out cards

Cloudiness (1–3)

A Drop's Journey · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 10–12 · Cut-out cards

Mass of soil

Ages 10–12 · Cut-out cards

Height of the back edge

Ages 10–12 · Cut-out cards

Pouring speed and start time

Ages 10–12 · Hint



The one thing we change is the independent variable; what we measure is measured; what we keep the same is constant.

Ages 10–12 · Question card

Team A poured the water on the bare tray in 15 seconds and on the covered tray in 60 seconds. Is this comparison fair?

A. Not fair: the pouring speed also changed, so two things differ.

B. Fair: both had 100 mL of water.

Ages 10–12 · Hint



How many things changed?

A Drop's Journey · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 10–12 · Question card

Team B pressed down the soil on the covered tray but not on the bare tray. What do you say?

A. Not fair; the soil must be prepared the same way on both trays.

B. It does not matter; only the cover matters.

Ages 10–12 · Hint



Could how tightly the soil is packed affect how water passes?

Ages 10–12 · Question card

Whatever the class result is, which sentence limits the conclusion correctly?

A. "On these trays and with this protocol, the cover affected the collected water in this direction."

B. "Straw prevents erosion on all fields."

C. "One team's result is enough."

Ages 10–12 · Hint



Where and at what scale was the experiment done?

Answers and explanations (teacher only)

Ages 7–9

1A	Amount of soil	Fits	Must be the same.
1B	Amount of water (100 mL)	Fits	Must be the same.
1C	Slope of the tray	Fits	Must be the same.
1D	Pouring the water at the same time	Fits	Must be the same.
1E	The cover on top of the soil	Does not fit	This is the one thing we change; if it were the same, there would be nothing to compare.
1F	Waiting time (3 minutes)	Fits	Must be the same.
2A	In the bare soil cup	Prediction: every option is accepted; only the real experiment shows the result.	You recorded your prediction. Only the real experiment will show the result.
2B	In the covered soil cup	Prediction: every option is accepted; only the real experiment shows the result.	You recorded your prediction. Only the real experiment will show the result.
2C	The same	Prediction: every option is accepted; only the real experiment shows the result.	You recorded your prediction. Only the real experiment will show the result.
4A	That is fine. I write my prediction and the result separately.	Fits	Yes. Keeping prediction and result apart is part of science.
4B	I change the result to match my prediction.	Does not fit	If we change the result, we cannot learn anything from the experiment.

Ages 10–12

1	Cover (bare / dry straw)	What we change	Independent variable: the only thing that differs between the trays.
1	Water collected in the cup (mL)	What we measure	Measured variable.
1	Cloudiness (1–3)	What we measure	Measured variable: with the READ card.
1	Mass of soil	What we keep the same	Kept constant.
1	Height of the back edge	What we keep the same	Kept constant.

1	Pouring speed and start time	What we keep the same	Kept constant: identical cups, at the same time.
2A	Not fair: the pouring speed also changed, so two things differ.	Fits	We cannot tell whether a difference comes from the cover or the rain speed.
2B	Fair: both had 100 mL of water.	Does not fit	The amount is the same but the speed is not; speed can also affect runoff.
3A	Not fair; the soil must be prepared the same way on both trays.	Fits	Pressing became a second variable.
3B	It does not matter; only the cover matters.	Does not fit	Pressed soil can let water through differently; that is why it is kept the same.
4A	"On these trays and with this protocol, the cover affected the collected water in this direction."	Fits	It keeps the conclusion at the scale of the experiment. The link between organic cover and water entering the soil is described by FAO (K12); our tray looks at it on a small scale.
4B	"Straw prevents erosion on all fields."	Does not fit	A small tray experiment does not give a certain result for all fields.
4C	"One team's result is enough."	Does not fit	Teams' results must be read together; one measurement may be due to chance.

A Drop's Journey: In a fair comparison only one thing changes. A prediction is an idea; only the real experiment shows the result. A small tray does not give a certain result for all fields.